Josue Godeme

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Education

Tsinghua University, Schwarzman College *(Beijing, China)* **Incoming August 2026 – Expected July 2027**
Master of Global Affairs, Schwarzman Scholar

Dartmouth College *(Hanover, NH)* **September 2022 – Expected June 2026**
Bachelor of Arts, Major in Computer Science, Minor in Chinese, Minor in International Studies **GPA: 3.79**
Awards: King Scholar (1/7 of 28K+), Stamps Scholar (1/7 of 400K+), Bosworth Award for Excellence in International Affairs (1/4)
Exchange Semesters: Beijing Normal University (Chinese Studies, 2025); AIT Budapest (Computer Science, 2024)
Leadership: Co-President, National Society of Black Engineers and CoderDojo; Executive Board, African Student Association

Experience

Bylaw *(stealth; making AI agents follow the law)* **July 2026 – Present**
Research Engineer Intern

- Translate written regulation into enforceable policies a machine can check, working from official statute text with every rule traceable back to the provision it came from.
- Found that simple allow-or-deny rules cannot capture ongoing compliance duties, which require tracking obligations, aggregating activity over time, and meeting deadlines; specified the extra layer needed to carry them.
- Built the checker that catches hallucinated legal citations in AI-generated text, verifying each against real source documents and never accusing without positive evidence: across 3,431 citations from published court opinions, the intuitive rule would have wrongly flagged 421 real ones.

Parsimmon *(document-parsing SaaS, competitor to Reducto)* **June 2026 – July 2026**
Software Engineering Intern

- Built the Parsimmon SaaS product end to end: multi-tenant FastAPI and TanStack Start application, public API endpoints, API-key management, org-scoped auth, and Stripe ledger-based billing, shipped as 13 reviewed PRs on a protected repo.
- Took the parsing engine from a local tool to a cloud service on AWS, behind a CI accuracy gate holding extraction quality at 91.7% before any merge.

Homy Robotics *(voice AI for senior-care companion robots)* **April 2026 – June 2026**
AI Engineering Intern

- Built the full voice system for a companion robot that talks with seniors in their homes: real-time audio, a swappable speech-to-text, LLM and text-to-speech pipeline, and a 30-table backend.
- Fixed the two failures that had kept the robot from ever speaking: a 30-minute install on its low-memory tablet, and audio sent to an output the hardware does not have. Cut install to 2 minutes and delivered its first working conversation.
- Built the robot memory layer: reviewed durable facts paired with semantic recall of past visits, isolated per household, where retrieval speaks a bridging phrase so the conversation never stalls while it looks something up.

Echo AI *(speech-first language learning application)* **September 2024 – Present**
Founder & Lead Developer

- Founded and built Echo AI end to end (speech recognition, synthesis, personalized feedback), growing to 100+ active users across multiple countries; raised $5,000 in grants.

Dartmouth College, Department of Computer Science *(Hanover, NH)* **Spring 2026**
Teaching Assistant, Computational Linguistics (CS72) and Introduction to Computer Science (CS1)

Research

Dartmouth Stamps Scholars Program *(Hanover, NH)* **June 2024 – Present**
Selected Student Researcher

- Built the first-ever machine translation, speech recognition, and speech synthesis systems for Adja, an extremely low-resource West African language; achieved BLEU 22, beating OpenAI and Google baselines.
- Led creation of the first digital corpus of 10,000 French-Adja sentence pairs with 10+ local translators; published at LREC 2026 as first author; secured a $10,000 Davis Projects for Peace grant.

Publications

- *Godeme, J., et al. (2025). Artificial Allies: Validation of Synthetic Text for Peer Support Tools through Data Augmentation in NLP Model Development.* ***Proceedings of the Pacific Symposium on Biocomputing****,* Hawaii, USA. (first author)
- *Godeme, J., et al. (2026). Adja-French Parallel Corpus: A New Resource for Machine Translation of a West African Under-Resourced Language.* ***LREC 2026*** (accepted; first author).

Skills

Languages & Tools: Python, TypeScript, Kotlin, PyTorch, Hugging Face, FastAPI, Next.js, React Native/Expo, Docker, AWS
AI & Voice: LLM agent design, tool calling, memory and retrieval, fine-tuning, ASR/TTS, LiveKit/WebRTC
Spoken: French (native), English (fluent), Chinese (proficient), Adja

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Selected Engineering Projects

Goloco / 1849.ai *(AI agent marketplace and settlement)* **2026**

TypeScript, TanStack Start, Solidity, MCP | 1849.ai

- Built marketplace and settlement infrastructure for AI agents, covering agent identities, service listings, task offers, delivery review, and payment receipts.
- Implemented an internal-credit ledger and USDC escrow on Base; verified agent-to-agent credit settlement locally and the full post-to-withdraw escrow cycle on the Base Sepolia testnet.
- Exposed task and delivery workflows through REST, an SDK, a CLI, and MCP; scoped connections to individual agents and enforced owner-defined spending limits.
- Added file deliveries up to 25 MB with direct storage uploads, checksum receipts, and bounded buyer previews, plus operator analytics for task completion and repeat hiring.

Relay *(Continuity and handoffs for coding agents)* **2026**

TypeScript, Bun, Git, Claude Code, Codex | GitHub

- Built an open-source CLI that runs coding agents and moves work between Claude Code and Codex while preserving the repository, task plan, and decisions.
- Implemented Git checkpoints that preserve staged and unstaged work without changing the user's branch or index; reconciled outgoing handoff notes against repository state and verification results.
- Added secret scanning, permission checks, crash recovery, and rollback behavior so a failed switch retains the saved work and leaves a recoverable handoff.
- Released compiled command-line binaries for macOS and Linux; validated a real Claude Code-to-Codex handoff in the early preview.

Crow Labs *(Experimental platform for AI shopping behavior)* **2026**

Python, FastAPI, Next.js, AWS | *Research platform*

- Built an experiment platform that freezes shopping tasks, randomizes product conditions, runs pinned model snapshots, and records every response, cost, and failure.
- Implemented resumable runs, immutable manifests, append-only JSONL evidence, and reproducible analysis, with artifact inventories and provenance linking measurements to their source records.
- Built an Observatory dashboard, Environment Studio, and Experiment Builder to configure campaigns, compare runs, and inspect prompts and results; checked hidden-field leakage and paired-condition isolation.
- Ran calibration and evidence-conflict pilots with Qwen2.5 0.5B and 1.5B, retaining failed controls and separating observed results from scientific interpretation.

Scribe *(Voice-controlled desktop text editor)* **2026**

Electron, TypeScript, React, Deepgram, OpenRouter | *Desktop prototypes*

- Built two Electron prototypes combining streaming dictation, spoken editing commands, and text-to-speech read-back with conventional typing and mouse interaction.
- Separated model-interpreted commands from deterministic text targeting: resolved edits to anchored spans, surfaced ambiguous matches for clarification, and applied changes through undoable transactions.
- Implemented punctuation and capitalization cleanup with a content-word preservation check, bounded repairs for speech-recognition errors, and keyboard navigation across words, sentences, and paragraphs.
- Integrated real and fake speech and model providers behind shared interfaces; added golden-trace benchmarks and tests for document behavior, renderer interactions, and end-to-end editing flows.